

Astro HW 11

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11.1

a)

11.2 Spiral galaxies - Tracing the Matter

a)

Given...

$$v_{esc} = \sqrt{\frac{2GM}{R}}$$
$$v_{esc} = 500 \text{ km s}^{-1} \quad 8.5 \text{ kpc}$$

We find that ...

$$v_{esc} = \frac{\sqrt{2GM}}{\sqrt{R}}$$
$$\frac{v_{esc}\sqrt{R}}{\sqrt{2G}} = \sqrt{M}$$
$$M = 2.470 \times 10^{11} \text{ M}_{\odot}$$

b)

i)

Given ...

$$M_V = -2.76 \log_{10} \left(\frac{P}{10 \text{ d}} \right) - 4.16$$
$$m_V = 18.9, \quad P = 31.47 \text{ d}$$
$$m - M = 5 \log_{10} \left(\frac{d}{10 \text{ pc}} \right) \rightarrow d = 10 \exp \left(\frac{m - M + 5}{5} \right) \text{ pc}$$

We find ...

$M_v = -5.534, \quad d = 7.706 \times 10^{-1} \text{ Mpc}$
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ii)

53.9 km/s

iii)

$$g = 3.914 \times 10^{-13} \text{ km/s}$$

iv)

Given ...

$$g = GM/R^2$$

$$g_{M31} = GM_{m31}/R^2, \quad g_{MW} = GM_{MW}/R^2$$

$$R = 7.706 \times 10^{-1} \text{ Mpc}$$

We can find ...

$$g_{M31} = 2g_{MW}$$

$$3.914 \times 10^{-13} = 2GM_{MW}/R^2$$

$$\frac{(3.914 \times 10^{-13})R^2}{2G} = M_{MW}$$

$$M_{MW} = 8.339 \times 10^{11} \text{ M}_{\odot}$$

11.3

a)

Given ...

$$R = D\theta$$

$$D = 16.8 \text{ Mpc}, \quad \theta = .1''$$

We find ...

$$R = 8.144 \text{ pc}$$

b)

From figure ...

$$-.1'' = 1350 \text{ km/s}$$

$$+.1'' = 650 \text{ km/s}$$

We find ...

$$c_{vel} = 350 \text{ km/s}$$

c)

Given ...

Assuming circular orbit ...

$$v^2 = \frac{GM}{R}$$

$$M = \frac{v^2 R}{G}$$

$$R = 8.144 \text{ pc}$$

We find ...

$$M_{bh} = 4.613 \times 10^{38} \text{ kg} = 2.320 \times 10^8 \text{ M}_{\odot}$$

11.4 Redshifts and the Hubble Law

a)

Given ...

$$Z = \frac{\lambda_{obsv} - \lambda_{rest}}{\lambda_{rest}}$$

$$\lambda_{rest} = 486.135 \text{ nm}, \quad \lambda_{obsv} = 497.97 \text{ nm}$$

We find ...

$$Z = .0023$$

b)

Given ...

$$m - M = 5 \log(d/10\text{pc})$$

$$v = H_0 d, \quad z \simeq \frac{v}{c}, \quad \rightarrow \frac{cz}{H_0} = d$$

$$H_0 = 70 \text{ km/s/Mpc}$$

With the goal of deriving ...

$$m - M = 5 \log(z) + 43.2$$

We can do ...

$$m - M = 5 \log_{10} \left(\frac{d}{10 \text{ pc}} \right)$$

$$m - M = 5 \log_{10} \left(\frac{\frac{cz}{H_0}}{10 \text{ pc}} \right)$$

$$m - M = 5 \log_{10} \left(\frac{\frac{cz}{H_0}}{10^{-5} \text{ Mpc}} \right), \quad m - M = 5 \log_{10} \left(\frac{cz}{H_0} \frac{10^5}{\text{Mpc}} \right)$$

$$m - M = 5 \log_{10} \left(\frac{c}{H_0} \frac{10^5}{\text{Mpc}} \right) + 5 \log_{10}(z)$$

$$5 \log_{10} \left(\frac{c}{H_0} \frac{10^5}{\text{Mpc}} \right) \approx 43.15$$

Hence ...

$m - M = 5 \log_{10}(z) + 43.2$
